

Mengfan Long

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PROFESSIONAL SUMMARY

- **Statistics and Econometrics graduate** with expertise in **quantitative finance**, **statistics**, and **computer science**. Skilled in **research and development**, bridging **financial modeling** such as **Copula**, **machine learning** like **Reinforcement Learning**, and **high-performance engineering** with **C++ trading system design**.

EDUCATION

University of Michigan

Ann Arbor, MI

Master of Applied Statistics

Expected May 2026

- **Coursework:** Time Series Analysis, Machine Learning, Probability Theory, Statistical Inference, Biostatistical Computing, Database Application Design, Regression Analysis, Data Structures.

University of Queensland

Brisbane, QLD

Bachelor of Economics in Econometrics (Honors)

Graduated Nov. 2023

- **Honors:** First Class Honours, Dean's Honor Roll (Top 10%), FitzGerald Scholarship.

PROFESSIONAL EXPERIENCE

Research Assistant

Jan 2025 – Present

Policy Analytical Lab (PAL)

Ann Arbor, Michigan

- Performed **financial policy analysis** on international regulatory systems, focusing on **capital adequacy**, **insurance solvency**, and **macroprudential frameworks**; supported **stress testing** and **risk analytics** for institutional decision-making.
- Developed **causal inference** and **predictive models** for policy evaluation using **propensity score matching**, **synthetic control**, and **time series decomposition**, contributing to **risk forecasting** and **capital planning**.

Quantitative Trading Intern

Jul – Sep 2022

Fx Trading Plus

Sydney, Australia

- Designed and fine-tuned **statistical** and **deep learning models** (XGBoost, LSTM) for **short-term price prediction**, improving signal precision and boosting **Sharpe ratio** by 10%.
- Developed and trained **reinforcement learning agents** (PPO, DDPG) in **FinRL environments** to optimize **portfolio rebalancing** under volatile market regimes, leveraging **reward shaping** and **policy iteration**.
- Engineered **transformer-based trading systems** for **dynamic position sizing**; deployed latency-efficient inference in **C++** with **PyTorch bindings**, improving **risk-adjusted returns** by 12%.

Finance Analyst Intern

Nov – Jan 2022

Luso International Bank

Guangzhou China

- Performed **equity and credit valuation analysis** using **DCF**, **comparable company analysis (CCA)**, and **precedent transactions**; produced pitch materials supporting buy-side investment decisions across **TMT** and **consumer sectors**.
- Built dynamic **financial models** in **Excel**, including **3-statement forecasting**, sensitivity analysis, and scenario testing; automated valuation dashboards integrating with **Power BI** and **Bloomberg** exports.
- Collaborated with **investment managers** and **research analysts** to synthesize **market trends** and **financial data** into clear, action-oriented **investment memos**. Delivered weekly presentations to **senior leadership**, demonstrating strong **communication**, **data storytelling**, and **time management** skills.

SKILLS SUMMARY

- **Computer Science & Programming:** Proficient in **Python**, **C++**, **R**, **SQL**, and **MATLAB**, with experience in **data structures**, **algorithms**, **OOP**, **multithreading**, **API design**, **Git**, **Linux**, and **Docker**.
- **Economics & Finance:** Strong understanding of **macroeconomics** and **microeconomics**. Skilled in **financial mathematics**, **portfolio optimization**, and **risk management**.
- **Mathematics Statistics:** Linear algebra, Optimization, Numerical methods, stochastic processes, Analysis, Copula modeling, Bayesian inference, Time series, Probability theory and Machine learning.